

# Hands on DAY-34 - Keerthivasan R

## Problem Statement

Many small organizations struggle to efficiently manage employee records such as personal details, department assignments, salary information, and job status. Traditional systems often rely on manual data handling or basic CRUD operations without advanced filtering, sorting, and searching capabilities.

There is a need for a dynamic web application that not only performs CRUD operations but also demonstrates powerful data querying capabilities using **LINQ (Language Integrated Query)** within **ASP.NET Core MVC**.

The goal of this project is to build a web-based Employee Management System that leverages LINQ for efficient data retrieval, filtering, grouping, sorting, and aggregation.

- Performs CRUD operations on employee records
  - Uses LINQ for advanced querying
  - Implements filtering, sorting, grouping, and aggregation
  - Demonstrates real-world data manipulation using Entity Framework Core
- 

## System Requirements

The application should allow users to:

### 1 Employee Management

- Add new employee records
- Edit employee details
- Delete employees
- View employee list

### 2 LINQ-Based Functionalities

- Search employees by name
  - Filter employees by department
  - Sort employees by salary (ascending/descending)
  - Group employees by department
  - Calculate:
    - Total number of employees
    - Average salary per department
    - Highest paid employee
  - Display employees hired within a specific date range
-

# Technical Constraints

- Use **ASP.NET Core MVC**
  - Use **Entity Framework Core (Code First or Database First)**
  - Perform all data queries using **LINQ**
  - Use SQL Server as the database
  - Apply proper model validation
  - Implement Dependency Injection
- 

## Expected Outcome

The final system should:

- Demonstrate practical use of LINQ in real-world scenarios
  - Efficiently handle data operations
  - Provide a clean and user-friendly interface
  - Follow MVC architecture principles
- 

### CODE:

 Model/ Employee.cs

```
using System;
using System.ComponentModel.DataAnnotations;

namespace EmployeeManagementSystem.Models
{
    public class Employee
    {
        [Key]
        public int Id { get; set; }

        [Required(ErrorMessage = "Name is required")]
        [StringLength(100)]
        public string Name { get; set; } = null!;

        [Required(ErrorMessage = "Email is required")]
        [EmailAddress]
        public string? Email { get; set; }

        [Required(ErrorMessage = "Department is required")]
        public string? Department { get; set; }

        [Required(ErrorMessage = "Salary is required")]
        [Range(1000, 500000, ErrorMessage = "Salary must be between 1000 and 500000")]
        public decimal Salary { get; set; }

        [Required(ErrorMessage = "Job Status is required")]
        public string? JobStatus { get; set; } // Active, Inactive, On Leave

        [Required(ErrorMessage = "Hire Date is required")]
        [DataType(DataType.Date)]
    }
}
```

```

        public DateTime HireDate { get; set; }

        public string? PhoneNumber { get; set; }

        public string? Address { get; set; }
    }
}

```

Model/ DepartmentGroup.cs

```

using System.Collections.Generic;

namespace EmployeeManagementSystem.Models
{
    public class DepartmentGroup
    {
        public string? Department { get; set; }
        public int EmployeeCount { get; set; }
        public decimal AverageSalary { get; set; }
        public List<Employee> ?Employees { get; set; }
    }
}

```

Model/ ErrorViewModel.cs

```

namespace EmployeeManagementSystem.Models
{
    public class ErrorViewModel
    {
        public string? RequestId { get; set; }

        public bool ShowRequestId => !string.IsNullOrEmpty(RequestId);
    }
}

```

Model/ StatisticsViewModel.cs

```

namespace EmployeeManagementSystem.Models
{
    public class StatisticsViewModel
    {
        public int TotalEmployees { get; set; }
        public decimal OverallAverageSalary { get; set; }
        public Employee? HighestPaidEmployee { get; set; }
        public List<DepartmentGroup>? DepartmentGroups { get; set; }
    }
}

```

Views/ Employees/ Index.cshtml

```

@model IEnumerable<EmployeeManagementSystem.Models.Employee>

@{
    ViewData["Title"] = "Employee Management";
}

<div class="container mt-4">
    <h1 class="mb-4">Employee Management System</h1>

    <!-- Success Message -->
    @if (TempData["Success"] != null)

```

```

    {
        <div class="alert alert-success alert-dismissible fade show"
role="alert">
            @TempData["Success"]
            <button type="button" class="btn-close" data-bs-
dismiss="alert"></button>
        </div>
    }

    <!-- LINQ Filtering and Search Section -->
    <div class="card mb-4">
        <div class="card-header bg-primary text-white">
            <h5>🔍 Search & Filter Employees (LINQ Queries)</h5>
        </div>
        <div class="card-body">
            <form method="get" asp-action="Index">
                <div class="row">
                    <div class="col-md-3">
                        <label>Search by Name:</label>
                        <input type="text" name="searchName" class="form-control"
value="@ViewBag.SearchName" placeholder="Enter name..." />
                    </div>
                    <div class="col-md-3">
                        <label>Filter by Department:</label>
                        <select name="departmentFilter" class="form-control">
                            <option value="">All Departments</option>
                            @foreach (var dept in ViewBag.Departments)
                            {
                                if (ViewBag.DepartmentFilter == dept)
                                {
                                    <option value="@dept" selected>@dept</option>
                                }
                                else
                                {
                                    <option value="@dept">@dept</option>
                                }
                            }
                        </select>
                    </div>
                    <div class="col-md-2">
                        <label>Hire Date From:</label>
                        <input type="date" name="fromDate" class="form-control"
value="@ViewBag.FromDate" />
                    </div>
                    <div class="col-md-2">
                        <label>Hire Date To:</label>
                        <input type="date" name="toDate" class="form-control"
value="@ViewBag.ToDate" />
                    </div>
                    <div class="col-md-2">
                        <label>&nbsp;</label>
                        <button type="submit" class="btn btn-primary form-
control">Apply Filters</button>
                    </div>
                </div>
            </form>
        </div>
    </div>

    <!-- Sorting and Actions -->
    <div class="row mb-3">
        <div class="col-md-6">
            <a asp-action="Create" class="btn btn-success">➕ Add New
Employee</a>
            <a asp-action="Statistics" class="btn btn-info">📊 View
Statistics</a>
        </div>
    </div>

```

```

        <div class="col-md-6 text-end">
            <a asp-action="Index" asp-route-sortOrder="asc" class="btn btn-outline-secondary">🔍 Sort by Salary (Low to High)</a>
            <a asp-action="Index" asp-route-sortOrder="desc" class="btn btn-outline-secondary">🔍 Sort by Salary (High to Low)</a>
        </div>
    </div>

    <!-- Employee List Table -->
    <div class="card">
        <div class="card-header">
            <h5>Employee Records (@Model.Count() employees found)</h5>
        </div>
        <div class="card-body">
            <table class="table table-striped table-hover">
                <thead>
                    <tr>
                        <th>ID</th>
                        <th>Name</th>
                        <th>Email</th>
                        <th>Department</th>
                        <th>Salary</th>
                        <th>Job Status</th>
                        <th>Hire Date</th>
                        <th>Actions</th>
                    </tr>
                </thead>
                <tbody>
                    @foreach (var item in Model)
                    {
                        <tr>
                            <td>@item.Id</td>
                            <td>@item.Name</td>
                            <td>@item.Email</td>
                            <td>@item.Department</td>
                            <td>@item.Salary.ToString("C")</td>
                            <td>
                                @if (item.JobStatus == "Active")
                                {
                                    <span class="badge bg-success">Active</span>
                                }
                                else if (item.JobStatus == "Inactive")
                                {
                                    <span class="badge bg-danger">Inactive</span>
                                }
                                else
                                {
                                    <span class="badge bg-warning">On
Leave</span>
                                }
                            </td>
                            <td>@item.HireDate.ToString("yyyy-MM-dd")</td>
                            <td>
                                <a asp-action="Edit" asp-route-id="@item.Id"
class="btn btn-sm btn-warning">Edit</a>
                                <a asp-action="Details" asp-route-id="@item.Id"
class="btn btn-sm btn-info">Details</a>
                                <a asp-action="Delete" asp-route-id="@item.Id"
class="btn btn-sm btn-danger">Delete</a>
                            </td>
                        </tr>
                    }
                </tbody>
            </table>
        </div>
    </div>
</div>

```



```

        <div class="form-group mb-3">
            <label asp-for="Salary" class="control-
label"></label>
            <input asp-for="Salary" class="form-control"
/>
            <span asp-validation-for="Salary"
class="text-danger"></span>
        </div>
    </div>
</div>
<div class="row">
    <div class="col-md-6">
        <div class="form-group mb-3">
            <label asp-for="JobStatus" class="control-
label"></label>
            <select asp-for="JobStatus" class="form-
control">
                <option value="Active">Active</option>
                <option
value="Inactive">Inactive</option>
                <option value="On Leave">On
Leave</option>
            </select>
            <span asp-validation-for="JobStatus"
class="text-danger"></span>
        </div>
    </div>
    <div class="col-md-6">
        <div class="form-group mb-3">
            <label asp-for="HireDate" class="control-
label"></label>
            <input asp-for="HireDate" type="date"
class="form-control" />
            <span asp-validation-for="HireDate"
class="text-danger"></span>
        </div>
    </div>
</div>
<div class="form-group mb-3">
    <label asp-for="PhoneNumber" class="control-
label"></label>
    <input asp-for="PhoneNumber" class="form-control" />
    <span asp-validation-for="PhoneNumber" class="text-
danger"></span>
</div>
<div class="form-group mb-3">
    <label asp-for="Address" class="control-
label"></label>
    <textarea asp-for="Address" class="form-control"
rows="3"></textarea>
    <span asp-validation-for="Address" class="text-
danger"></span>
</div>
<div class="form-group">
    <button type="submit" class="btn btn-success">Create
Employee</button>
    <a asp-action="Index" class="btn btn-
secondary">Cancel</a>
</div>
</form>
</div>
</div>
</div>

```

```

    </div>
</div>

```

```

@section Scripts {
    @{
        await Html.RenderPartialAsync("_ValidationScriptsPartial");
    }
}

```

Views/ Employees/ Delete.cshtml

```

@model EmployeeManagementSystem.Models.Employee

```

```

@{
    ViewData["Title"] = "Delete Employee";
}

```

```

<div class="container mt-4">
    <div class="row justify-content-center">
        <div class="col-md-8">
            <div class="card">
                <div class="card-header bg-danger text-white">
                    <h3>Delete Employee</h3>
                </div>
                <div class="card-body">
                    <div class="alert alert-warning">
                        <strong>Warning!</strong> Are you sure you want to delete
this employee? This action cannot be undone.
                    </div>

                    <dl class="row">
                        <dt class="col-sm-3">Name</dt>
                        <dd class="col-sm-9">@Model.Name</dd>

                        <dt class="col-sm-3">Email</dt>
                        <dd class="col-sm-9">@Model.Email</dd>

                        <dt class="col-sm-3">Department</dt>
                        <dd class="col-sm-9">@Model.Department</dd>

                        <dt class="col-sm-3">Salary</dt>
                        <dd class="col-sm-9">@Model.Salary.ToString("C")</dd>

                        <dt class="col-sm-3">Job Status</dt>
                        <dd class="col-sm-9">@Model.JobStatus</dd>

                        <dt class="col-sm-3">Hire Date</dt>
                        <dd class="col-sm-9">@Model.HireDate.ToString("yyyy-MM-
dd")</dd>

                        <dt class="col-sm-3">Phone Number</dt>
                        <dd class="col-sm-9">@Model.PhoneNumber</dd>

                        <dt class="col-sm-3">Address</dt>
                        <dd class="col-sm-9">@Model.Address</dd>
                    </dl>

                    <form asp-action="Delete">
                        <input type="hidden" asp-for="Id" />
                        <button type="submit" class="btn btn-danger">Confirm
Delete</button>
                        <a asp-action="Index" class="btn btn-
secondary">Cancel</a>
                    </form>
                </div>
            </div>
        </div>
    </div>
</div>

```



```

        </div>
    </div>
</div>

```

Views/ Employees/ Details.cshtml

```

@model EmployeeManagementSystem.Models.Employee

@{
    ViewData["Title"] = "Employee Details";
}

<div class="container mt-4">
    <div class="row justify-content-center">
        <div class="col-md-8">
            <div class="card">
                <div class="card-header bg-info text-white">
                    <h3>Employee Details</h3>
                </div>
                <div class="card-body">
                    <dl class="row">
                        <dt class="col-sm-3">ID</dt>
                        <dd class="col-sm-9">@Model.Id</dd>

                        <dt class="col-sm-3">Name</dt>
                        <dd class="col-sm-9">@Model.Name</dd>

                        <dt class="col-sm-3">Email</dt>
                        <dd class="col-sm-9">@Model.Email</dd>

                        <dt class="col-sm-3">Department</dt>
                        <dd class="col-sm-9">@Model.Department</dd>

                        <dt class="col-sm-3">Salary</dt>
                        <dd class="col-sm-9">@Model.Salary.ToString("C")</dd>

                        <dt class="col-sm-3">Job Status</dt>
                        <dd class="col-sm-9">@Model.JobStatus</dd>

                        <dt class="col-sm-3">Hire Date</dt>
                        <dd class="col-sm-9">@Model.HireDate.ToString("yyyy-MM-
dd")</dd>

                        <dt class="col-sm-3">Phone Number</dt>
                        <dd class="col-sm-9">@Model.PhoneNumber</dd>

                        <dt class="col-sm-3">Address</dt>
                        <dd class="col-sm-9">@Model.Address</dd>
                    </dl>

                    <div class="form-group">
                        <a asp-action="Edit" asp-route-id="@Model.Id" class="btn
btn-warning">Edit</a>
                        <a asp-action="Index" class="btn btn-secondary">Back to
List</a>
                    </div>
                </div>
            </div>
        </div>
    </div>
</div>

```

Views/ Employees/ Edit.cshtml

```
@model EmployeeManagementSystem.Models.Employee

@{
    ViewData["Title"] = "Edit Employee";
}

<div class="container mt-4">
    <div class="row justify-content-center">
        <div class="col-md-8">
            <div class="card">
                <div class="card-header bg-warning">
                    <h3>Edit Employee</h3>
                </div>
                <div class="card-body">
                    <form asp-action="Edit">
                        <div asp-validation-summary="ModelOnly" class="text-
danger"></div>

                        <input type="hidden" asp-for="Id" />

                        <div class="form-group mb-3">
                            <label asp-for="Name" class="control-label"></label>
                            <input asp-for="Name" class="form-control" />
                            <span asp-validation-for="Name" class="text-
danger"></span>
                        </div>

                        <div class="form-group mb-3">
                            <label asp-for="Email" class="control-label"></label>
                            <input asp-for="Email" class="form-control" />
                            <span asp-validation-for="Email" class="text-
danger"></span>
                        </div>

                        <div class="row">
                            <div class="col-md-6">
                                <div class="form-group mb-3">
                                    <label asp-for="Department" class="control-
label"></label>

                                    <select asp-for="Department" class="form-
control">

                                        <option value="IT">IT</option>
                                        <option value="HR">HR</option>
                                        <option value="Finance">Finance</option>
                                        <option
value="Marketing">Marketing</option>
                                        <option value="Sales">Sales</option>
                                    </select>
                                    <span asp-validation-for="Department"
class="text-danger"></span>
                                </div>
                            </div>
                            <div class="col-md-6">
                                <div class="form-group mb-3">
                                    <label asp-for="Salary" class="control-
label"></label>

                                    <input asp-for="Salary" class="form-control"
/>

                                    <span asp-validation-for="Salary"
class="text-danger"></span>
                                </div>
                            </div>
                        </div>

                        <div class="row">
                            <div class="col-md-6">
                                <div class="form-group mb-3">
                                    <label asp-for="Title" class="control-label"></label>

                                    <input asp-for="Title" class="form-control" />
                                    <span asp-validation-for="Title" class="text-
danger"></span>
                                </div>
                            </div>
                            <div class="col-md-6">
                                <div class="form-group mb-3">
                                    <label asp-for="Phone" class="control-label"></label>

                                    <input asp-for="Phone" class="form-control" />
                                    <span asp-validation-for="Phone" class="text-
danger"></span>
                                </div>
                            </div>
                        </div>
                    </form>
                </div>
            </div>
        </div>
    </div>
</div>
```

```

        <label asp-for="JobStatus" class="control-
label"></label>
        <select asp-for="JobStatus" class="form-
control">
            <option value="Active">Active</option>
            <option
value="Inactive">Inactive</option>
            <option value="On Leave">On
Leave</option>
        </select>
        <span asp-validation-for="JobStatus"
class="text-danger"></span>
    </div>
</div>
<div class="col-md-6">
    <div class="form-group mb-3">
        <label asp-for="HireDate" class="control-
label"></label>
        <input asp-for="HireDate" type="date"
class="form-control" />
        <span asp-validation-for="HireDate"
class="text-danger"></span>
    </div>
</div>
</div>
<div class="form-group mb-3">
    <label asp-for="PhoneNumber" class="control-
label"></label>
    <input asp-for="PhoneNumber" class="form-control" />
    <span asp-validation-for="PhoneNumber" class="text-
danger"></span>
</div>
<div class="form-group mb-3">
    <label asp-for="Address" class="control-
label"></label>
    <textarea asp-for="Address" class="form-control"
rows="3"></textarea>
    <span asp-validation-for="Address" class="text-
danger"></span>
</div>
<div class="form-group">
    <button type="submit" class="btn btn-warning">Save
Changes</button>
    <a asp-action="Index" class="btn btn-
secondary">Cancel</a>
</div>
</form>
</div>
</div>
</div>
</div>
</div>
</div>

```

```

@section Scripts {
    @{
        await Html.RenderPartialAsync("_ValidationScriptsPartial");
    }
}

```

Views/ Employees/ Statistics.cshtml

```
@model EmployeeManagementSystem.Models.StatisticsViewModel
```

```

@{
    ViewData["Title"] = "Employee Statistics";
}

<div class="container mt-4">
    <h1 class="mb-4"><img alt="Bar chart icon" data-bbox="318 138 338 153"/> Employee Statistics Dashboard</h1>

    @if (Model == null)
    {
        <div class="alert alert-danger">Error: Statistics data could not be
loaded.</div>
    }
    else
    {
        <div class="row">
            <div class="col-md-4">
                <div class="card text-white bg-primary mb-3">
                    <div class="card-header">Total Employees</div>
                    <div class="card-body">
                        <h1 class="card-title display-
4">@Model.TotalEmployees</h1>
                        <p class="card-text">Total active employees in system</p>
                    </div>
                </div>
            </div>

            <div class="col-md-4">
                <div class="card text-white bg-success mb-3">
                    <div class="card-header">Average Salary (Overall)</div>
                    <div class="card-body">
                        <h1 class="card-title display-
4">@Model.OverallAverageSalary.ToString("C0")</h1>
                        <p class="card-text">Company-wide average salary</p>
                    </div>
                </div>
            </div>

            <div class="col-md-4">
                <div class="card text-white bg-warning mb-3">
                    <div class="card-header">Highest Paid Employee</div>
                    <div class="card-body">
                        @* Use the Null-Conditional Operator '?' to prevent
crashes *@
                        @if (Model.HighestPaidEmployee != null)
                        {
                            <h4 class="card-
title">@Model.HighestPaidEmployee.Name</h4>
                            <p class="card-text">Salary:
@Model.HighestPaidEmployee.Salary.ToString("C")</p>
                            <small>Department:
@Model.HighestPaidEmployee.Department</small>
                        }
                        else
                        {
                            <p class="card-text">No employees found</p>
                        }
                    </div>
                </div>
            </div>
        </div>

        <div class="card mt-4">
            <div class="card-header bg-info text-white">
                <h5><img alt="Pie chart icon" data-bbox="308 903 328 918"/> Department-wise Statistics (LINQ Grouping &
Aggregation)</h5>
            </div>
        </div>
    }
}

```

```

<div class="card-body">
    @* Check if DepartmentGroups exists and has any data *@
    @if (Model.DepartmentGroups != null &&
Model.DepartmentGroups.Any())
    {
        @foreach (var dept in Model.DepartmentGroups)
        {
            <div class="card mb-3">
                <div class="card-header bg-secondary text-white">
                    <h5>Department: @dept.Department</h5>
                </div>
                <div class="card-body">
                    <div class="row">
                        <div class="col-md-4">
                            <strong>👤 Employee Count:</strong>
@dept.EmployeeCount
                        </div>
                        <div class="col-md-4">
                            <strong>💰 Average Salary:</strong>
@dept.AverageSalary.ToString("C")
                        </div>
                        <div class="col-md-4">
                            @* Added '?' check before Sum to ensure
the list isn't null *@
                            <strong>💰 Total Salary Pool:</strong>
@(dept.Employees?.Sum(e => e.Salary).ToString("C") ?? "$0.00")
                        </div>
                    </div>
                    <hr />
                    <h6>Employees in @dept.Department:</h6>
                    <table class="table table-sm">
                        <thead>
<tr><th>Name</th><th>Salary</th><th>Status</th><th>Hire Date</th></tr>
                        </thead>
                        <tbody>
                            @if (dept.Employees != null)
                            {
                                @foreach (var emp in dept.Employees)
                                {
                                    <tr>
                                        <td>@emp.Name</td>
                                        <td>@emp.Salary.ToString("C")</td>
                                        <td>@emp.JobStatus</td>
                                        <td>@emp.HireDate.ToString("yyyy-MM-dd")</td>
                                    </tr>
                                }
                            }
                        </tbody>
                    </table>
                </div>
            </div>
        }
    }
    else
    {
        <p class="text-center">No departmental data available.</p>
    }
</div>
</div>
<div class="mt-3">
    <a asp-action="Index" class="btn btn-primary">< Back to Employee List</a>
</div>

```

</div>

📁 Controllers/ EmployeeController.cs

```
using Microsoft.AspNetCore.Mvc;
using Microsoft.EntityFrameworkCore;
using EmployeeManagementSystem.Models;
using EmployeeManagementSystem.Data;
using System;
using System.Linq;

namespace EmployeeManagementSystem.Controllers
{
    public class EmployeesController : Controller
    {
        private readonly ApplicationDbContext _context;

        // Dependency Injection
        public EmployeesController(ApplicationDbContext context)
        {
            _context = context;
        }

        // GET: Employees - Main view with all LINQ features
        public async Task<IActionResult> Index(
            string searchName,
            string departmentFilter,
            string sortOrder,
            DateTime? fromDate,
            DateTime? toDate)
        {
            // Start with all employees - LINQ Query
            var employees = _context.Employees.AsQueryable();

            // ①SEARCH by name using LINQ
            if (!string.IsNullOrEmpty(searchName))
            {
                employees = employees.Where(e => e.Name.Contains(searchName));
                ViewBag.SearchName = searchName;
            }

            // ②FILTER by department using LINQ
            if (!string.IsNullOrEmpty(departmentFilter))
            {
                employees = employees.Where(e => e.Department ==
departmentFilter);
                ViewBag.DepartmentFilter = departmentFilter;
            }

            // ③FILTER by date range using LINQ
            if (fromDate.HasValue)
            {
                employees = employees.Where(e => e.HireDate >= fromDate.Value);
                ViewBag.FromDate = fromDate.Value.ToString("yyyy-MM-dd");
            }
            if (toDate.HasValue)
            {
                employees = employees.Where(e => e.HireDate <= toDate.Value);
                ViewBag.ToDate = toDate.Value.ToString("yyyy-MM-dd");
            }

            // ④SORT by salary using LINQ
            ViewBag.SalarySortParam = string.IsNullOrEmpty(sortOrder) ?
"salary_desc" : "";
        }
    }
}
```

```

ViewBag.CurrentSort = sortOrder;

switch (sortOrder)
{
    case "salary_desc":
        employees = employees.OrderByDescending(e => e.Salary);
        break;
    default:
        employees = employees.OrderBy(e => e.Salary);
        break;
}

// Get unique departments for filter dropdown
var departments = await _context.Employees
    .Select(e => e.Department)
    .Distinct()
    .ToListAsync();
ViewBag.Departments = departments;

var employeeList = await employees.ToListAsync();

return View(employeeList);
}

// GET: Employees/Statistics - LINQ Aggregation
public async Task<IActionResult> Statistics()
{
    var employees = await _context.Employees.ToListAsync();

    // 5 GROUP by department and calculate aggregates using LINQ
    var departmentGroups = employees
        .GroupBy(e => e.Department)
        .Select(g => new DepartmentGroup
        {
            Department = g.Key,
            EmployeeCount = g.Count(),
            AverageSalary = g.Average(e => e.Salary),
            Employees = g.ToList()
        })
        .ToList();

    // Calculate overall statistics using LINQ
    var statistics = new StatisticsViewModel
    {
        TotalEmployees = employees.Count(),
        OverallAverageSalary = employees.Any() ? employees.Average(e =>
e.Salary) : 0,
        HighestPaidEmployee = employees.OrderByDescending(e =>
e.Salary).FirstOrDefault(),
        DepartmentGroups = departmentGroups
    };

    return View(statistics);
}

// GET: Employees/Details/5
public async Task<IActionResult> Details(int? id)
{
    if (id == null)
    {
        return NotFound();
    }

    var employee = await _context.Employees
        .FirstOrDefaultAsync(m => m.Id == id);

    if (employee == null)

```

```

        {
            return NotFound();
        }

        return View(employee);
    }

    // GET: Employees/Create
    public IActionResult Create()
    {
        return View();
    }

    // POST: Employees/Create
    [HttpPost]
    [ValidateAntiForgeryToken]
    public async Task<IActionResult>
Create([Bind("Name,Email,Department,Salary,JobStatus,HireDate,PhoneNumber,Address
")] Employee employee)
    {
        if (ModelState.IsValid)
        {
            _context.Add(employee);
            await _context.SaveChangesAsync();
            TempData["Success"] = "Employee created successfully!";
            return RedirectToAction(nameof(Index));
        }
        return View(employee);
    }

    // GET: Employees/Edit/5
    public async Task<IActionResult> Edit(int? id)
    {
        if (id == null)
        {
            return NotFound();
        }

        var employee = await _context.Employees.FindAsync(id);
        if (employee == null)
        {
            return NotFound();
        }
        return View(employee);
    }

    // POST: Employees/Edit/5
    [HttpPost]
    [ValidateAntiForgeryToken]
    public async Task<IActionResult> Edit(int id,
[Bind("Id,Name,Email,Department,Salary,JobStatus,HireDate,PhoneNumber,Address")]
Employee employee)
    {
        if (id != employee.Id)
        {
            return NotFound();
        }

        if (ModelState.IsValid)
        {
            try
            {
                _context.Update(employee);
                await _context.SaveChangesAsync();
                TempData["Success"] = "Employee updated successfully!";
            }
            catch (DbUpdateConcurrencyException)

```



```

        {
            if (!EmployeeExists(employee.Id))
            {
                return NotFound();
            }
            else
            {
                throw;
            }
        }
        return RedirectToAction(nameof(Index));
    }
    return View(employee);
}

// GET: Employees/Delete/5
public async Task<IActionResult> Delete(int? id)
{
    if (id == null)
    {
        return NotFound();
    }

    var employee = await _context.Employees
        .FirstOrDefaultAsync(m => m.Id == id);
    if (employee == null)
    {
        return NotFound();
    }

    return View(employee);
}

// POST: Employees/Delete/5
[HttpPost, ActionName("Delete")]
[ValidateAntiForgeryToken]
public async Task<IActionResult> DeleteConfirmed(int id)
{
    var employee = await _context.Employees.FindAsync(id);
    if (employee != null)
    {
        _context.Employees.Remove(employee);
        await _context.SaveChangesAsync();
        TempData["Success"] = "Employee deleted successfully!";
    }

    return RedirectToAction(nameof(Index));
}

private bool EmployeeExists(int id)
{
    return _context.Employees.Any(e => e.Id == id);
}
}
}

```

#### Data/ ApplicationDbContext.cs

```

using Microsoft.EntityFrameworkCore;
using EmployeeManagementSystem.Models;
namespace EmployeeManagementSystem.Data
{
    public class ApplicationDbContext : DbContext
    {
        public ApplicationDbContext(DbContextOptions<ApplicationDbContext>
options)

```

```

        : base(options)
    {
    }
    public DbSet<Employee> Employees { get; set; }

    protected override void OnModelCreating(ModelBuilder modelBuilder)
    {
        base.OnModelCreating(modelBuilder);

        // Seed some initial data
        modelBuilder.Entity<Employee>().HasData(
            new Employee
            {
                Id = 1,
                Name = "John Doe",
                Email = "john@company.com",
                Department = "IT",
                Salary = 75000,
                JobStatus = "Active",
                HireDate = new DateTime(2023, 1, 15),
                PhoneNumber = "1234567890",
                Address = "123 Main St"
            },
            new Employee
            {
                Id = 2,
                Name = "Jane Smith",
                Email = "jane@company.com",
                Department = "HR",
                Salary = 65000,
                JobStatus = "Active",
                HireDate = new DateTime(2023, 3, 20),
                PhoneNumber = "0987654321",
                Address = "456 Oak Ave"
            },
            new Employee
            {
                Id = 3,
                Name = "Bob Johnson",
                Email = "bob@company.com",
                Department = "IT",
                Salary = 85000,
                JobStatus = "Active",
                HireDate = new DateTime(2022, 6, 10),
                PhoneNumber = "5555555555",
                Address = "789 Pine Rd"
            }
        );
    }
}

```

OUTPUT:

Employee Management System

Search & Filter Employees (LINQ Queries)

Search by Name:  
Enter name...

Filter by Department:  
All Departments

Hire Date From:  
dd-mm-yyyy

Hire Date To:  
dd-mm-yyyy

Apply Filters

Add New Employee

View Statistics

Sort by Salary (Low to High)

Sort by Salary (High to Low)

Employee Records (4 employees found)

| ID | Name        | Email            | Department | Salary      | Job Status | Hire Date  | Actions                                  |
|----|-------------|------------------|------------|-------------|------------|------------|------------------------------------------|
| 4  | Sanjai      | sanjai@gmail.com | HR         | ₹ 50,000.00 | Active     | 2026-04-02 | <a>Edit</a> <a>Details</a> <a>Delete</a> |
| 2  | Jane Smith  | jane@company.com | HR         | ₹ 65,000.00 | Active     | 2023-03-20 | <a>Edit</a> <a>Details</a> <a>Delete</a> |
| 1  | John Doe    | john@company.com | IT         | ₹ 75,000.00 | Active     | 2023-01-15 | <a>Edit</a> <a>Details</a> <a>Delete</a> |
| 3  | Bob Johnson | bob@company.com  | IT         | ₹ 85,000.00 | Active     | 2022-06-10 | <a>Edit</a> <a>Details</a> <a>Delete</a> |

Employee Statistics Dashboard

Total Employees  
4  
Total active employees in system

Average Salary (Overall)  
₹ 68,750  
Company-wide average salary

Active Job Openings  
500 Job Openings  
Openings: ₹ 1,00,00,000  
Openings: ₹ 1

Department-wise Statistics (LINQ Grouping & Aggregation)

Department: IT

Employee Count: 2

Average Salary: ₹ 80,000.00

Total Salary Pool: ₹ 1,60,000.00

Employees in IT:

| Name        | Salary      | Status | Hire Date  |
|-------------|-------------|--------|------------|
| John Doe    | ₹ 75,000.00 | Active | 2023-01-15 |
| Bob Johnson | ₹ 85,000.00 | Active | 2022-06-10 |

Department: HR

Employee Count: 2

Average Salary: ₹ 57,500.00

Total Salary Pool: ₹ 1,15,000.00

Employees in HR:

| Name       | Salary      | Status | Hire Date  |
|------------|-------------|--------|------------|
| Jane Smith | ₹ 65,000.00 | Active | 2023-03-20 |
| Sanjai     | ₹ 50,000.00 | Active | 2026-04-02 |

Back to Employees List

Add New Employee

Name  
Jaskirat

Email  
Singh@gmail.com

Department  
HR

Salary  
100000

JobStatus  
Active

HireDate  
02-04-2026

PhoneNumber  
8866677777

Address  
Lahore, Pak

Create Employee

Cancel

## Delete Employee

Warning! Are you sure you want to delete this employee? This action cannot be undone.

**Name** Sanjay  
**Email** sanjay@gmail.com  
**Department** HR  
**Salary** ₹ 50,000.00  
**Job Status** Active  
**Hire Date** 2005-04-02  
**Phone Number** 555552222  
**Address** New, Mumbai

Confirm Delete

Cancel

## Edit Employee

Name

Jane Smith

Email

jane@company.com

Department

HR

Salary

65000.00

Job Status

Active

Hire Date

20-03-2023

PhoneNumber

0987654321

Address

456 Oak Ave

Save Changes

Cancel